Valentin Machado

3D Web
Software Engineer

CONTACT

Lyon · Driving licence
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WEB 3D

three.js · Babylon.js · WebGL · iTowns · 3D Tiles · point clouds

WEB & BACK-END

JavaScript · Node.js · Express · Socket.IO · SQLite · Webpack · Web Audio · d3

OTHER

C++ · Java · Nim · Git · Docker · Linux · ffmpeg · Unity · Godot

LANGUAGES

French — native
English — B2 (Cambridge English)

INTERESTS

Music (guitar, bass, composition, music production in Ableton), sound design, cinema, drawing, squash, hiking.

> Software engineer with **8 years of experience** in 3D web applications: professional 3D editors, multiplayer games, geospatial data visualisation. I design and ship complete products, from WebGL rendering to the server.

EXPERIENCE

NodaLab — Founder, developer
nodalab.app — 2025 – present

Web application for learning music theory, designed and built on my own with a view to bringing it to market.

- Node graph editor (chords, rhythms, guitar fretboard, player), also rendered as an interactive page for the learner.
- In-browser audio synchronised with the display (Tone.js, MIDI, samples).
- Node.js / Express / SQLite back end: accounts, Google and GitHub OAuth, messaging, multilingual content; unit and end-to-end tests.

Université de Lyon — Research Engineer
Jan 2021 – Dec 2023

LabEx IMU: 3D web applications on urban data, with the VCity team at LIRIS.

- Architect of **ud-viz** (open source): multiplayer geospatial game engine, point cloud visualisation.
- **UD-IMUV**: multiplayer serious game about the city of Lyon (published scientific paper).
- **Galeri3**: collaborative application for geolocated discussions and 3D data import.
- Contributions to **iTowns** (IGN): 3D Tiles styling.

Freelance — Software Engineer
Apr 2018 – Jan 2021

- Small WebGL projects for clients on Fiverr.

Wanadev — Engineer, project manager
Mar 2016 – Apr 2018

- **Popup Builder** (three.js): 2D/3D design of kit houses, generating the manufacturing files for the factory.
- **SMAIO i-PLAN** (Babylon.js): spine editor built from a point cloud, for surgical planning.
- Project management and client relations; final-year project on global illumination in WebGL.

PERSONAL PROJECTS

Subtitle Game
subtitle-game.mache.lol — 2025 – 2026

Multiplayer game where players subtitle film clips: real-time synchronised video (Socket.IO), editing and export of the final film with ffmpeg.

Guess It
guess-it.mache.lol — 2026

Quiz generator (films, music, countries, painting…) fed by around ten public APIs (TMDb, Wikidata, Wikipedia…), with an SQLite cache and rate-limited request queues.

EDUCATION

2024 – 2026	A break for music and sound: preparation for the BTS in audiovisual production (sound option, French two-year vocational diploma), then a preparatory year at the ENM (music school) in Villeurbanne.
2012 – 2016	**CPE Lyon** — engineering degree (MSc level), computer science major, including an exchange at **Queen's University** (Kingston, Canada) in artificial intelligence and algorithms.
2009 – 2012	Classes préparatoires (MPSI, MP), intensive maths & physics programme, Lycée Alphonse Daudet, Nîmes.